WORKED SOLUTION

STEP III 2002, Question 7

Pure • Coordinate Geometry Identities

STEP III 2002, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q7

The key idea

The expression $s^2 + b^2 - a^2$ should look familiar: it is what the cosine rule produces in triangle OAB, where $OA = s$ and $OB = b$. So the first identity is really the opening result about $\cos^2\alpha + \cos^2\beta$, applied to the two angles that OB makes with the axes. After that, the second identity is pure algebra built on the first, and the inequality falls out because a sum of two squares is at least each square.

Solution

The angle result

Suppose first that $\alpha + \beta = \frac{1}{2}\pi$. Then $\cos\beta = \cos(\frac{1}{2}\pi - \alpha) = \sin\alpha$, so

$$\cos^2\alpha + \cos^2\beta = \cos^2\alpha + \sin^2\alpha = 1.$$

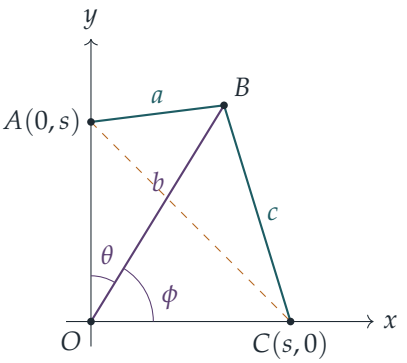
Conversely, suppose $\cos^2\alpha + \cos^2\beta = 1$. Then $\cos^2\beta = 1 - \cos^2\alpha = \sin^2\alpha$. Both α and β are acute, so $\cos\beta > 0$ and $\sin\alpha > 0$, and taking positive square roots gives

$$\cos\beta = \sin\alpha = \cos(\tfrac{1}{2}\pi - \alpha).$$

Now β and $\frac{1}{2}\pi - \alpha$ both lie in $(0, \frac{1}{2}\pi)$, where cosine is strictly decreasing and so takes each value only once. Hence $\beta = \frac{1}{2}\pi - \alpha$, that is, $\alpha + \beta = \frac{1}{2}\pi$. This proves the "if and only if".

The first identity

Let $\theta = \angle AOB$ and $\phi = \angle BOC$.



The cosine rule in triangle OAB (sides $OA = s$, $OB = b$, $AB = a$) and in triangle OCB (sides $OC = s$, $OB = b$, $CB = c$) gives

$$\cos\theta = \frac{s^2 + b^2 - a^2}{2sb}, \qquad \cos\phi = \frac{s^2 + b^2 - c^2}{2sb}.$$

Because B lies in the open first quadrant, the ray OB lies strictly between the positive y-axis (along OA) and the positive x-axis (along OC). So θ and ϕ are both acute and $\theta + \phi = \frac{1}{2}\pi$. By the "only if" direction above, $\cos^2\theta + \cos^2\phi = 1$, that is,

$$\frac{(s^2 + b^2 - a^2)^2}{4s^2b^2} + \frac{(s^2 + b^2 - c^2)^2}{4s^2b^2} = 1.$$

Multiplying by $4s^2b^2$ gives the first identity:

$$(s^2 + b^2 - a^2)^2 + (s^2 + b^2 - c^2)^2 = 4s^2b^2.$$

(For a direct check, write $B = (x, y)$. Then $a^2 = x^2 + (y - s)^2$, $b^2 = x^2 + y^2$ and $c^2 = (x - s)^2 + y^2$, so $s^2 + b^2 - a^2 = 2sy$ and $s^2 + b^2 - c^2 = 2sx$, and the left-hand side is $4s^2(x^2 + y^2) = 4s^2b^2$.)

The second identity

To keep the algebra tidy, write

$$P = s^2 + b^2 - a^2, \qquad Q = s^2 + b^2 - c^2, \qquad S = s^2 + b^2, \qquad D = s^2 - b^2,$$

so that $a^2 = S - P$ and $c^2 = S - Q$. The first identity says $P^2 + Q^2 = 4s^2b^2$. Since $4s^2b^2 = (s^2 + b^2)^2 - (s^2 - b^2)^2 = S^2 - D^2$, it can be rewritten as

$$D^2 = S^2 - P^2 - Q^2. \tag{†}$$

Now $a^2 + c^2 = 2S - P - Q$ and $2s^2 = S + D$, $2b^2 = S - D$, so

$$2s^2 - a^2 - c^2 = (P + Q - S) + D, \qquad 2b^2 - a^2 - c^2 = (P + Q - S) - D.$$

Ask yourself Why is this split into $(P+Q-S)\pm D$ worth making?

Because $(u+D)^2+(u-D)^2=2u^2+2D^2$: the cross terms cancel, and D then appears only as D^2, which (†) lets us replace by something involving P, Q and S alone. With $u=P+Q-S$,

$$\begin{aligned}&(2s^2-a^2-c^2)^2+(2b^2-a^2-c^2)^2\\&\quad=2(P+Q-S)^2+2D^2\\&\quad=2(P^2+Q^2+S^2+2PQ-2PS-2QS)+2(S^2-P^2-Q^2)\\&\quad=4S^2-4PS-4QS+4PQ\\&\quad=4(S-P)(S-Q)\\&\quad=4a^2c^2,\end{aligned}$$

using (†) in the second line. This is the second identity.

The inequality

Both squares on the left of the second identity are non-negative, so

$$(2b^2-a^2-c^2)^2\le 4a^2c^2,\qquad\text{so}\qquad |2b^2-a^2-c^2|\le 2ac,$$

taking non-negative square roots (note $a,c>0$). This says

$$-2ac\le 2b^2-a^2-c^2\le 2ac,$$

that is,

$$a^2-2ac+c^2\le 2b^2\le a^2+2ac+c^2,$$

which is $(a-c)^2\le 2b^2\le(a+c)^2$, as required.

Equality holds exactly when $2s^2=a^2+c^2$. Since $AC^2=2s^2$, by the converse of Pythagoras that means $\angle ABC$ is a right angle: B lies on the circle with diameter AC.

Checks

- Take $s=2$ and $B=(1,3)$. Then $a^2=2$, $b^2=c^2=10$. The first identity reads $12^2+4^2=160=4\cdot4\cdot10$; the second reads $(-4)^2+8^2=80=4\cdot2\cdot10$; and the inequality is $3.06\le 20\le 20.94$.
- The point $B=(s,s)$ is the fourth corner of the square, so $\angle ABC$ is right and $a=c=s$, $b=s\sqrt{2}$: then $2b^2=4s^2=(a+c)^2$, the equality case.
- A random test of all three results at 2×10^5 points B found no failures.

Common slips

- Leaving out the acute condition in the converse: without it, $\cos^2\beta=\sin^2\alpha$ does not force $\beta=\frac{1}{2}\pi-\alpha$ (for example, $\beta=\frac{1}{2}\pi+\alpha$ also works).

- Expanding the second identity in a, b, c, s by brute force without using the first. It can be done, but it is easy to drop a term; the substitution S, D keeps it under control.
- Taking square roots of $(2b^2 - a^2 - c^2)^2 \leq 4a^2c^2$ without the modulus, and so getting only one half of the inequality.

Beyond STEP

The quantity $2b^2 - a^2 - c^2$ is $2s(x + y - s)$, which is ± 4 times the area of triangle ABC, so the second identity is the cosine rule and the area formula $\frac{1}{2}ac \sin B$ combined into $\cos^2 B + \sin^2 B = 1$. The upper bound is also a case of Ptolemy's inequality: for any four points, $OB \cdot AC \leq OA \cdot BC + OC \cdot AB$, with equality for a cyclic quadrilateral, and here this reads $b\,s\sqrt{2} \leq s(a + c)$. Relations between distances like this lead on to the study of metric spaces and of distance geometry; Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.